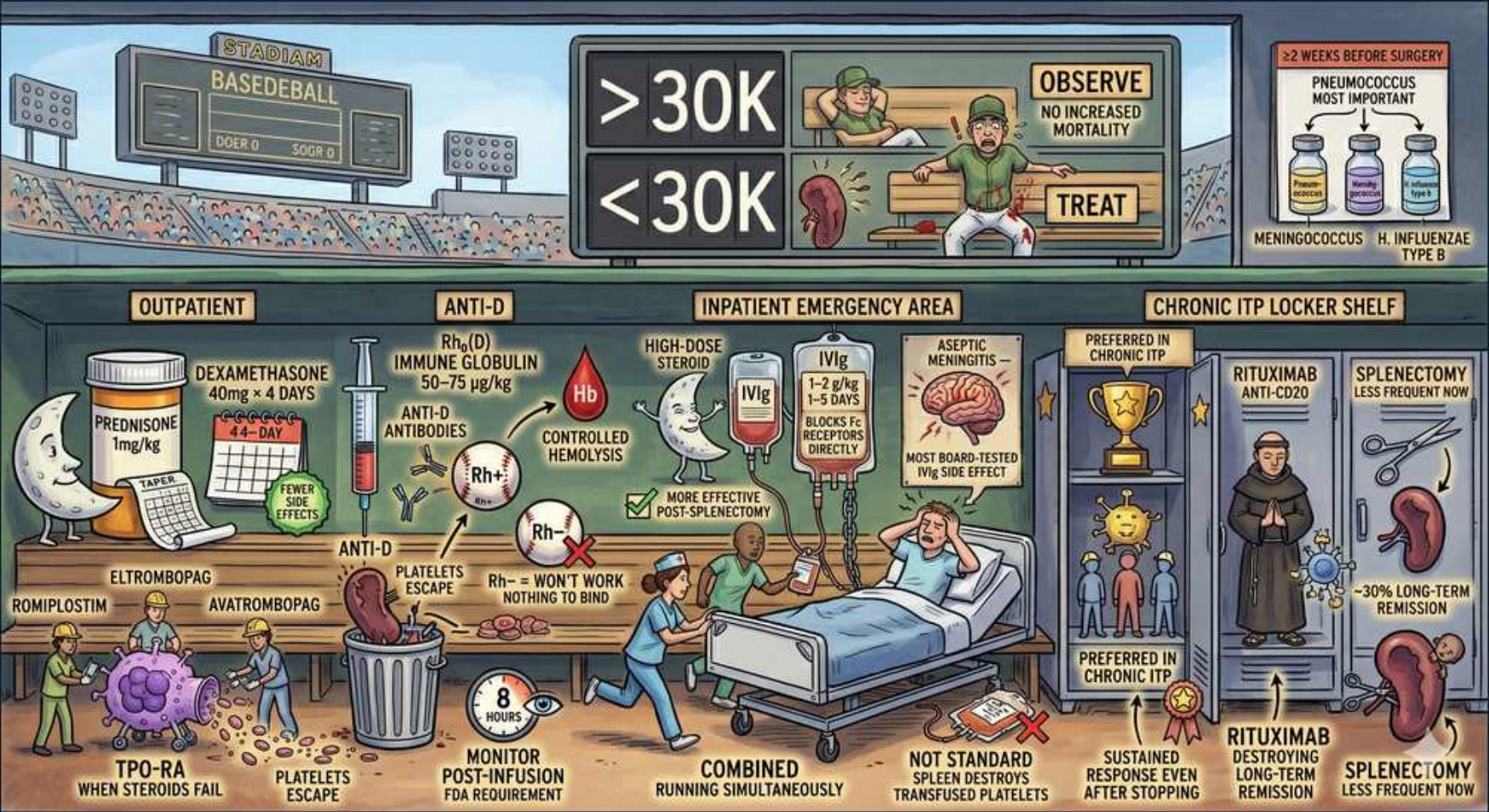


Platelet disorders — ITP treatment (Basedball treatment)





## 1. >30K observe

### WHAT YOU SEE

The stadium scoreboard reading '>30K' beside a relaxed player on a bench labeled OBSERVE / NO INCREASED MORTALITY — the high score means just watch the game, no treatment.

### WHAT IT ENCODES

If platelets are >30,000 AND there is no bleeding, the answer is OBSERVE — no treatment, because spontaneous serious bleeding is rare above this threshold and stable mild ITP carries no increased mortality. You treat the patient/bleeding, not the number. Counts of 30,000–50,000 with no bleeding are watched, often indefinitely.

### BOARD TRAP

The classic distractor is to start prednisone on every patient with a low platelet count. An asymptomatic adult with platelets of 40,000–60,000 needs OBSERVATION, not steroids. Treating the number alone (especially above 30,000) is the wrong management — exposure to steroid toxicity without benefit.

## 2. <30K treat

### WHAT YOU SEE

The scoreboard flipping to '<30K' with the player sent to the TREAT bench — a low count drops you below the threshold where you must act.

### WHAT IT ENCODES

Initiate treatment when platelets are <30,000 OR there is clinically significant bleeding at any count. The first-line agent for the non-bleeding patient is corticosteroids; add IVIG/anti-D when you need a fast rise. Procedures have their own thresholds (e.g., aim ~50,000 for surgery, ~80,000–100,000 for neurosurgery).

### BOARD TRAP

Don't withhold treatment just because the absolute count is 'only mildly low' if the patient is actively bleeding — significant bleeding is itself an indication to treat regardless of the number. The mirror trap is treating a stable count of 35,000 with no bleeding; the threshold for the non-bleeder is <30,000.

## 3. Vaccinate before splenectomy

### WHAT YOU SEE

A vaccination placard listing PNEUMOCOCCUS (MOST IMPORTANT), H. INFLUENZAE TYPE B, and MENINGOCOCCUS, stamped '2 WEEKS BEFORE SURGERY' — the encapsulated-organism shots given pre-splenectomy.

### WHAT IT ENCODES

Before an elective splenectomy, vaccinate against the ENCAPSULATED organisms — pneumococcus (most important), H. influenzae type b, and meningococcus — ideally ≥2 weeks pre-op to allow an immune response. This prevents overwhelming post-splenectomy infection (OPSI), a lifelong asplenia risk.

### BOARD TRAP

Forgetting the encapsulated-organism vaccines before splenectomy (or giving them the day of surgery instead of ≥2 weeks prior) is the planted error. Pneumococcus is the single most important. The wrong answer skips vaccination — asplenic patients face lethal OPSI from Strep pneumoniae, H. influenzae, and Neisseria meningitidis.

## 4. Glucocorticoids first-line

### WHAT YOU SEE

The crescent-moon PREDNISONE pill (40 mg x 4 days) with a dexamethasone bottle in the OUTPATIENT zone — the moon marks steroids as the first-line, go-to drug.

### WHAT IT ENCODES

FIRST-LINE pharmacologic therapy is corticosteroids: either prednisone ~1 mg/kg/day tapered over weeks, OR pulsed high-dose dexamethasone 40 mg/day × 4 days. They work by reducing splenic macrophage clearance and antibody production. Response takes days; most relapse on taper, which is why steroids are rarely a durable single solution.

### BOARD TRAP

Choosing splenectomy or rituximab as INITIAL therapy is wrong — those are second-line. Steroids are first-line for the stable, non-emergent patient. The other trap is expecting an instant response: steroids take a few days, so for a bleeding emergency you must ADD IVIG, not rely on steroids alone for speed.

## 5. IVIG / anti-D fast

### WHAT YOU SEE

Two red IVIG infusion bags labeled for a fast 1-2 g/kg rise — the drip bags are the rapid-rescue agents that work in days.

### WHAT IT ENCODES

For a RAPID platelet rise — active bleeding or before an urgent procedure — give IVIG (1 g/kg/day × 1–2 days) or anti-D, alongside steroids. IVIG works fast (24–72 h) by saturating splenic macrophage Fc receptors so they stop eating coated platelets. The effect is quick but transient (a few weeks).

### BOARD TRAP

Relying on IVIG/anti-D for DURABLE control is the trap — their effect is temporary (weeks), so they are bridges for emergencies or procedures, not maintenance therapy. Also don't substitute IVIG for source control in life-threatening bleeding: severe ICH-level bleeding needs steroids + IVIG + platelet transfusion together, not IVIG alone.

## 6. Anti-D Rh+ only

### WHAT YOU SEE

The ANTI-D figure tied to Rh+ blood with 'Rh- WON'T WORK / NOTHING TO BIND' crossed out — anti-D only works if there are Rh-positive red cells and a spleen.

### WHAT IT ENCODES

Anti-D (Rho) immunoglobulin only works in Rh(D)-POSITIVE patients who still HAVE a spleen. Mechanism: it coats the patient's own Rh+ red cells, and the spleen preferentially clears those antibody-coated RBCs instead of platelets — a controlled, mild hemolysis that 'distracts' the spleen. No spleen or Rh-negative = no effect.

### BOARD TRAP

Giving anti-D to an Rh-NEGATIVE patient or to someone who is post-SPLENECTOMY is the wrong answer — it cannot work without Rh+ red cells for the spleen to consume. Also watch for its boxed risk: rare severe intravascular hemolysis/DIC, so it is monitored. Use IVIG instead when anti-D's prerequisites are not met.

## 7. TPO-RA when steroids fail

### WHAT YOU SEE

Three figures labeled ROMIPLOSTIM, ELTROMBOPAG, and AVATROMBOPAG captioned 'TPO-RA WHEN STEROIDS FAIL' — the megakaryocyte-boosters used second-line.

### WHAT IT ENCODES

Thrombopoietin receptor agonists — eltrombopag and avatrombopag (oral) and romiplostim (subcutaneous) — stimulate megakaryocytes to make MORE platelets, directly countering the production-block arm of ITP. They are SECOND-LINE for chronic/persistent or steroid-refractory ITP and are often used long-term.

### BOARD TRAP

Two traps: (1) TPO-RAs are NOT first-line — don't pick them over steroids initially; (2) they raise thrombotic risk and eltrombopag is hepatotoxic and requires chelation-style spacing from polyvalent cations (calcium, antacids, dairy) that block its absorption. The wrong answer ignores these and co-administers eltrombopag with calcium/antacids.



## 8. High-dose steroid inpatient

### WHAT YOU SEE

The INPATIENT EMERGENCY bay showing HIGH-DOSE STEROID plus IVIG bags COMBINED / RUNNING SIMULTANEOUSLY over a patient in bed — everything at once for a severe bleed.

### WHAT IT ENCODES

For SEVERE or life-threatening bleeding (e.g., intracranial hemorrhage, count <10,000 with wet purpura), give COMBINATION therapy SIMULTANEOUSLY: high-dose IV corticosteroids (e.g., methylprednisolone) PLUS IVIG, and add platelet transfusion if life-threatening. Hitting multiple mechanisms at once gives the fastest, highest peak count.

### BOARD TRAP

Sequential, single-agent therapy in a crashing ITP bleed is the error — you do NOT start steroids and 'wait to see' before adding IVIG. Life-threatening hemorrhage demands all available modalities at once (steroids + IVIG ± platelets). Picking monotherapy for an ICH-level bleed costs time the patient does not have.

## 9. Platelet transfusion not standard

### WHAT YOU SEE

A bag of transfused platelets captioned 'NOT STANDARD / SPLEEN DESTROYS TRANSFUSED PLATELETS' — donor platelets get eaten, so transfusion is not routine.

### WHAT IT ENCODES

Platelet transfusion is NOT standard therapy in ITP because transfused platelets are rapidly opsonized and destroyed by the same autoantibody, so they barely raise the count. Reserve transfusion for LIFE-THREATENING bleeding only, and give it together with IVIG/steroids to blunt destruction.

### BOARD TRAP

Routinely transfusing platelets to 'fix the number' in stable ITP is the wrong answer — they are consumed within hours and accomplish little. Note the contrast with TTP, where platelets are CONTRAINDICATED (fuel thrombosis); in ITP they are merely ineffective/not-standard but ARE permitted as a rescue in life-threatening hemorrhage.

## 10. IVIG aseptic meningitis

### WHAT YOU SEE

A pained cartoon brain clutching its head labeled ASEPTIC MENINGITIS next to the IVIG bags — the headache-and-stiff-neck side effect of IVIG.

### WHAT IT ENCODES

The most board-tested adverse effect of IVIG is ASEPTIC MENINGITIS — severe headache, photophobia, neck stiffness, and a sterile CSF pleocytosis appearing within a day or two of infusion; it is self-limited. Other IVIG harms: volume overload, acute kidney injury, thrombosis, and (rarely) anaphylaxis in IgA-deficient patients.

### BOARD TRAP

Calling post-IVIG headache + meningismus a bacterial/infectious meningitis and starting empiric antibiotics is the trap — recognize the temporal link to the recent infusion. IVIG aseptic meningitis is sterile and resolves on its own; also remember to screen IgA-deficient patients before IVIG to avoid anaphylaxis.

## 11. Rituximab chronic

### WHAT YOU SEE

A hooded MONK figure labeled RITUXIMAB / ANTI-CD20, 'PREFERRED IN CHRONIC ITP' — the robed monk is the anti-CD20 B-cell killer for chronic disease.

### WHAT IT ENCODES

Rituximab, an anti-CD20 monoclonal antibody, depletes B cells and thereby reduces autoantibody production; it is a SECOND-LINE option preferred for chronic/refractory ITP, especially when wanting to avoid or defer splenectomy. Responses can be durable but are not universal.

### BOARD TRAP

Using rituximab as initial therapy is wrong — it is second-line. Board-relevant risks: reactivation of HEPATITIS B (screen HBsAg/anti-HBc and treat/prophylax before giving it) and the rare risk of PML. The trap answer gives rituximab without checking hepatitis B serologies first.

## 12. Splenectomy

### WHAT YOU SEE

A SPLENECTOMY figure captioned 'LESS FREQUENT NOW' with '~30% LONG-TERM REMISSION / SUSTAINED RESPONSE AFTER STOPPING' — surgical spleen removal for durable remission.

### WHAT IT ENCODES

Splenectomy removes both the main site of platelet destruction and the main antibody factory, producing SUSTAINED long-term remission in roughly two-thirds of patients (commonly cited ~30%+ durable cure-level remission). It is used LESS often now that TPO-RAs and rituximab exist, and is typically deferred ~6–12 months to allow spontaneous/medical remission.

### BOARD TRAP

Rushing to splenectomy as an early or first-line maneuver is the trap — modern practice defers it, exhausts medical options first, and reserves it for chronic refractory disease. And never forget the prerequisite: encapsulated-organism vaccination ≥2 weeks before surgery and lifelong OPSI awareness afterward.